



Mobile Application Testing

There are broadly 2 kinds of testing that take place on mobile devices:

Types of Testing in mobiles

1. Hardware Testing
2. Software/Application Testing

1. Hardware testing:

The device including the internal processors, internal hardware, screen sizes, resolution, space or memory, camera, radio, Bluetooth, WIFI etc. This is sometimes referred to as, simple “Mobile Testing”.

2. Software or Application testing:

The applications that work on mobile devices and their functionality is tested. It is called the “Mobile Application Testing”

Types of Mobile Applications

a) **Native apps:** created for use on a platform like mobile and tablets.

Image: Native Application



b) Mobile web apps

server-side apps to access website/s on mobile using different browsers like chrome, Firefox by connecting to a mobile network or wireless network like WIFI.

Image: Web Application



c) Hybrid apps

combinations of native app and web app. They run on devices or offline and are written using web technologies like HTML5 and CSS.



Need For Mobile Application

As with any application, Mobile application testing is also very important, as client is usually in millions for a certain product – and a product with bugs is never appreciated. It often results in

- monetary losses
- legal issue
- brand image damage.

Key Challenges in Mobile Application Testing

1. **Variety of Mobile Devices**- difference in screen sizes, input methods (QWERTY, touch, normal) with different hardware capabilities.
 - ❏ Device models change frequently as well.
 - ❏ Devices also vary by country.
 - ❏ Device detection: do you know what your users have?
2. **Diversity in Mobile Platforms/OS**- The major ones are Android, IOS, Symbian, Windows Phone, and BlackBerry .
3. **Mobile network operators**- like GSM and CDMA.
Frequent updates – (like android- 4.2, 4.3, 4.4, iOS-5.x, 6.x)
4. **Scripting**- The variety of devices makes executing the test script (Scripting) a key challenge. As devices differ in keystrokes, input methods, menu structure and display properties single , script does not function on every device.

Types of Mobile Application

To address all the above technical aspects, the following types of testing are performed on Mobile applications.

1. **Compatibility testing**
2. **Interface testing**- Testing of menu options, buttons, bookmarks, history, settings, and navigation flow of the application.
3. **Services testing**- Testing the services of the application online and offline.
4. **Low level resource testing**: Testing of memory usage, auto deletion of temporary files, local database known as low level resource testing.
5. **Installation tests**- Validation of the application by installing /uninstalling it on the devices.

6.Laboratory Testing- Laboratory testing, usually carried out by network carriers, is done by simulating the complete wireless network. This test is performed to find out any glitches when a mobile application uses voice and/or data connection to perform some functions.

7.Memory Leakage Testing- Memory leakage can result in poor performance of the application and the overall slowdown of the system. As mobile devices have significant constraints of available memory, memory leakage testing is crucial for the proper functioning of an application.

8.Interrupt Testing- An application while functioning may face several interruptions like incoming calls or network coverage outage and recovery. The different types of interruptions are:

- Incoming and Outgoing SMS and MMS

- Incoming and Outgoing calls

- Incoming Notifications

- Battery Removal

- Cable Insertion and Removal for data transfer

- Network outage and recovery

- Media Player on/off

- Device Power cycle

9- Certification Testing- To get a certificate of compliance, each mobile device needs to be tested against the guidelines set by different mobile platforms.

The Certified Mobile Application Tester popularly known as CMAT certification exam is offered by the Global Association for Quality Management (GAQM).

10. Usability testing- This is important as the usability of an application is its key to commercial success(it is nothing but user friendliness). make sure that the mobile app is easy to use and provides a satisfactory user experience to the customers

11. Functional Testing- Functional testing ensures that the application is working as per the requirements. Most of the test conducted for this is driven by the user interface and call flows.

12. Performance Testing---Testing of the application under certain conditions such as low battery, bad network coverage, low available memory, simultaneous access to application's server by several users and other conditions.

Two kinds of automation tools are available to test mobile apps:

1.Object based mobile testing tools- automation by mapping elements on the device screen into objects. This approach is independent of screen size and mainly used for Android devices.

Eg:- ranorex, jamo solution

2.Image based mobile testing tools- create automation scripts based on screen coordinates of elements.

Eg:- Sikuli, Egg Plant, RoutineBot

Test Cases for Testing a Mobile App

In addition to functionality based test cases, Mobile application testing requires special test cases which should cover following scenarios.

- Speed of the application**- the response time on different devices, with different memory parameters, with different network types etc.
- Data requirements** – For installation as well as to verify if the user with limited data plan will be able to download it.
- Functionality of the application**- make sure application is not crashing due to network failure or anything else.

Mobile-Application-Sample-Test-Cases.xls

testing tools

- FoneMonkey: iOS
- Apple xCode: Mac, iPhone and iPad
- Robotium: Android
- eggPlant: automation tool to work across all devices
- Perfecto Mobile
- Device Anywhere

THANK
YOU